A BRIEF OVERVIEW OF AI

A sector by sector analysis

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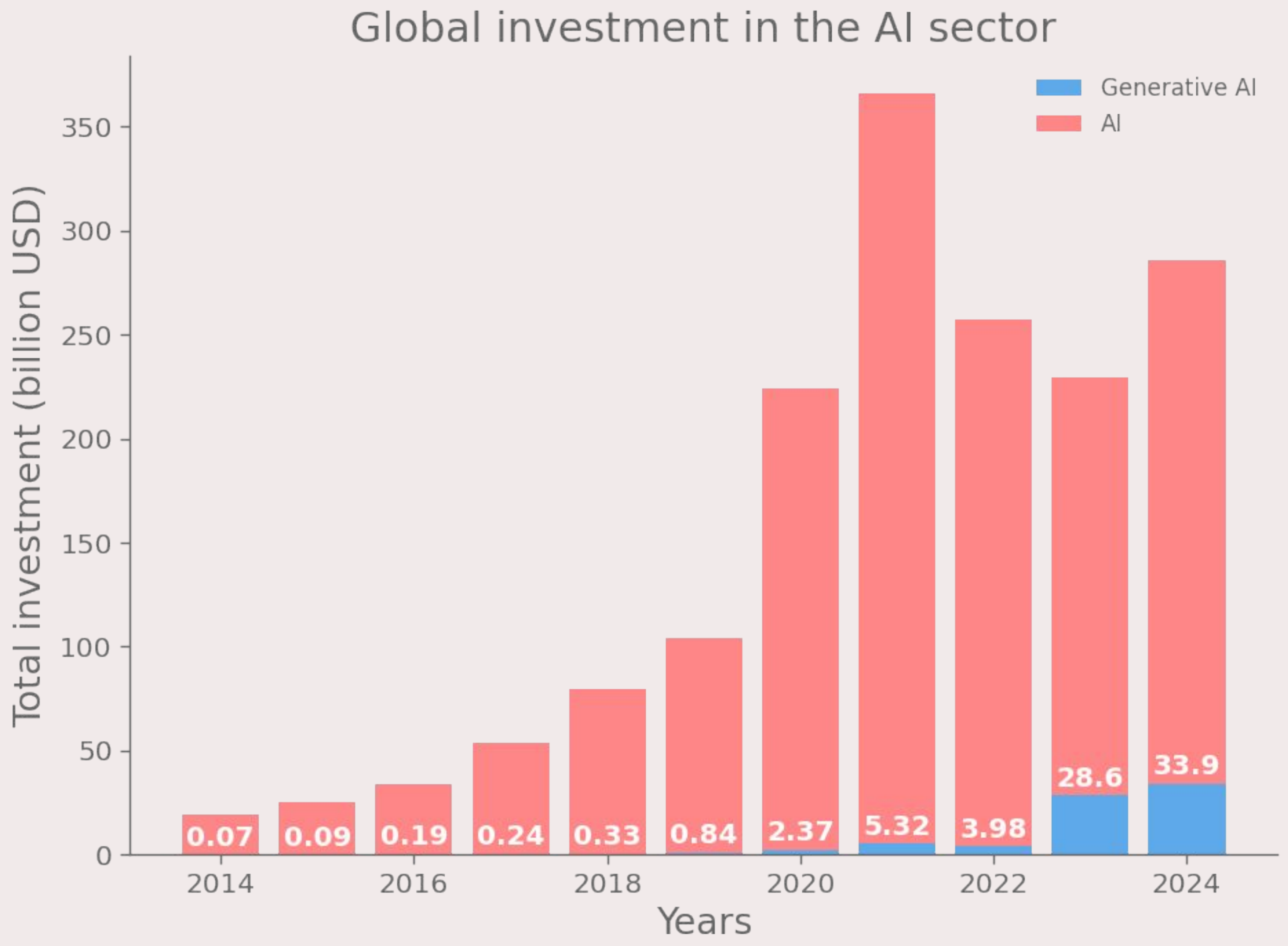
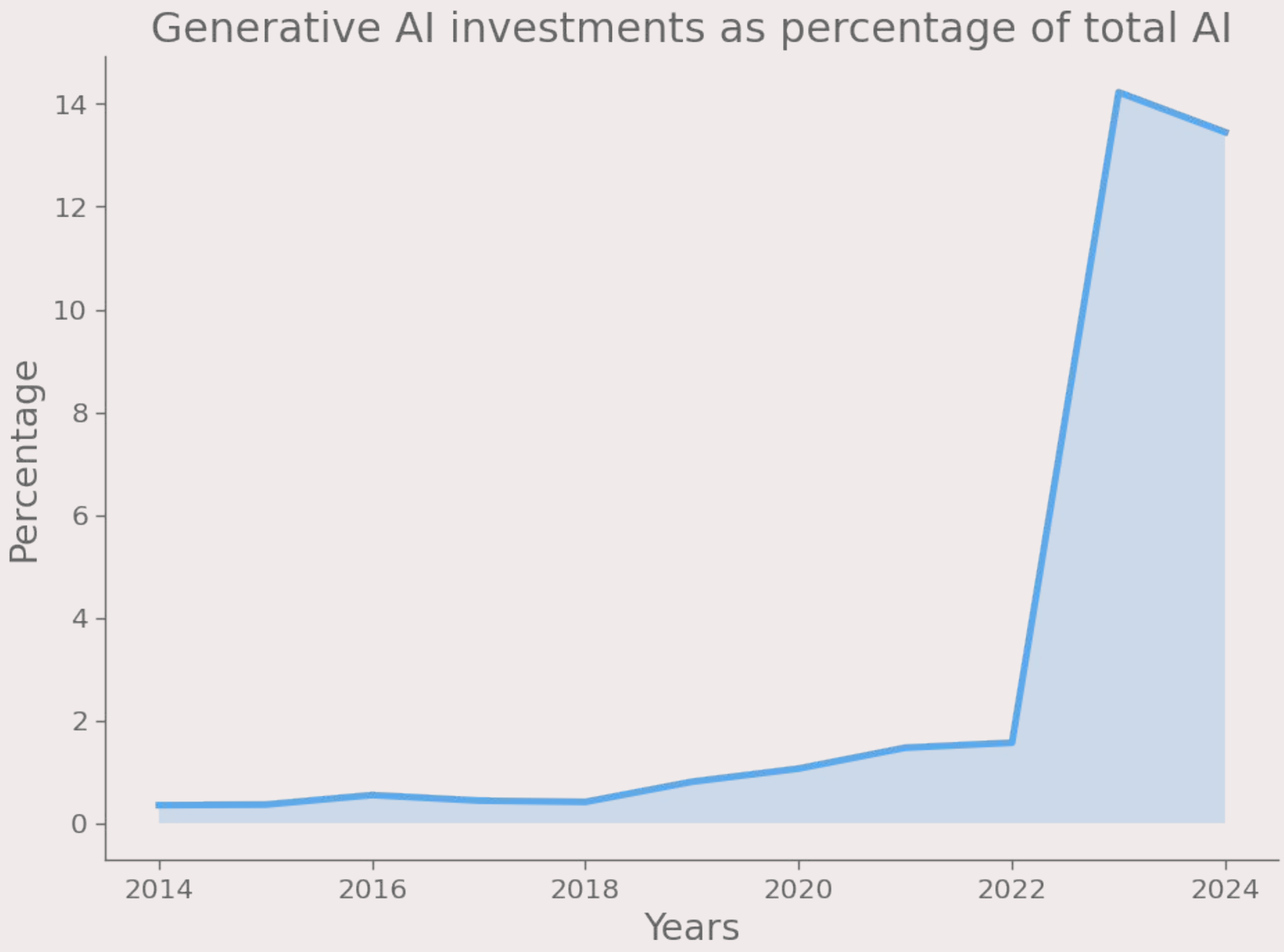
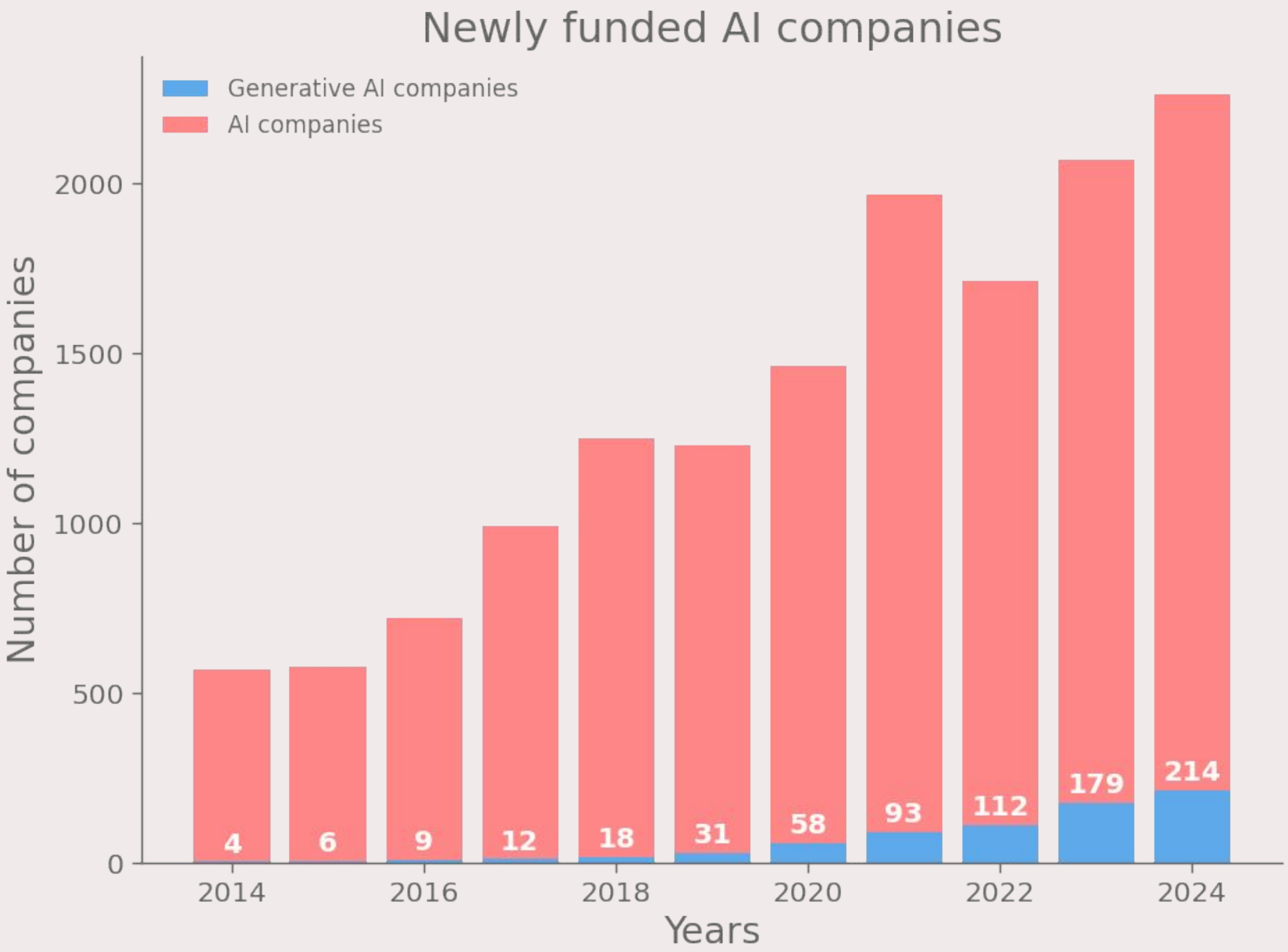
ECONOMY

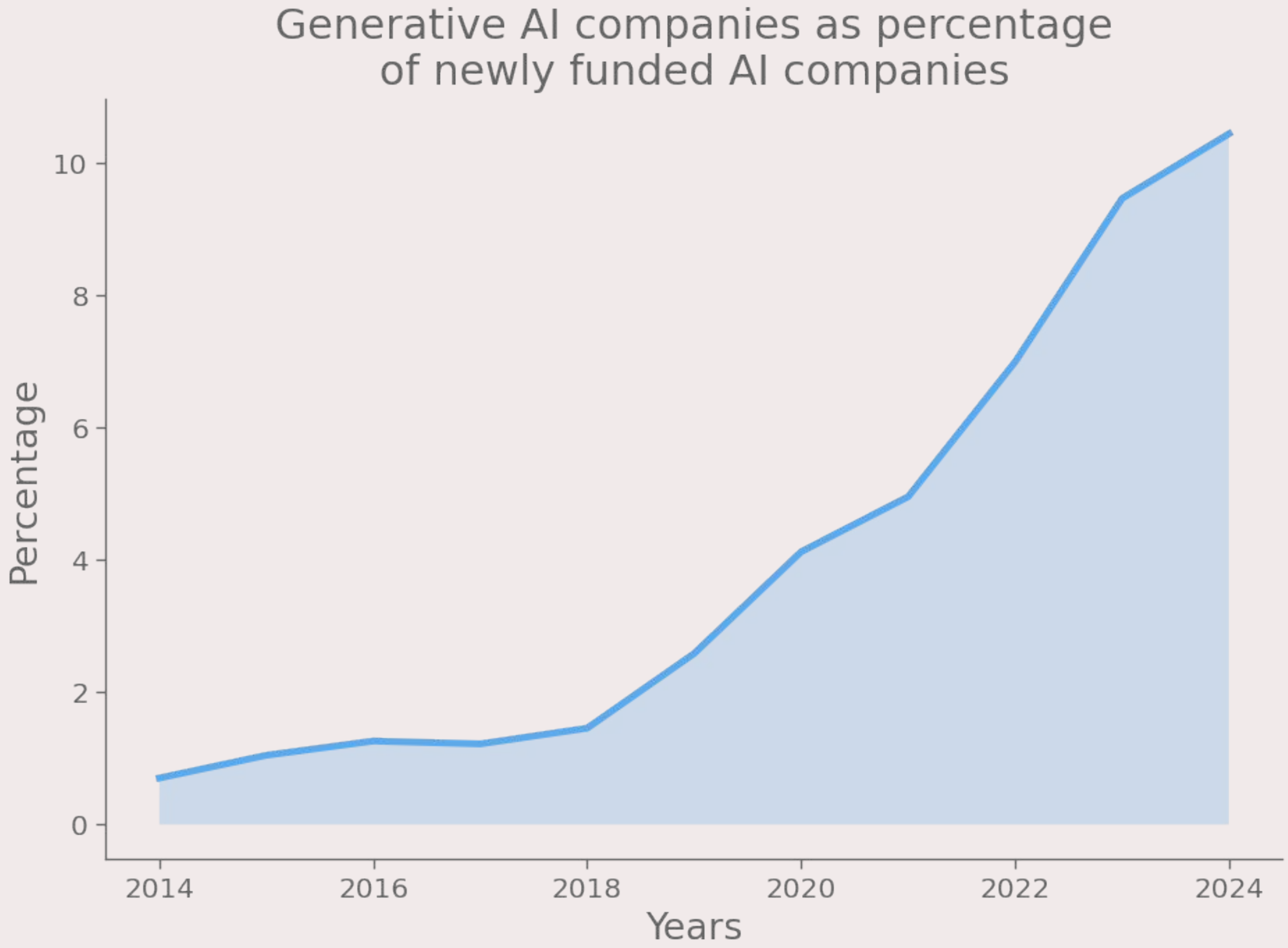

Global investment in the AI sector
Generative AI
AI
Total investment (billion USD)
Years
0
50
100
150
200
250
300
350
2014
2016
2018
2020
2022
2024
0.07
0.09
0.19
0.24
0.33
0.84
2.37
5.32
3.98
28.6
33.9

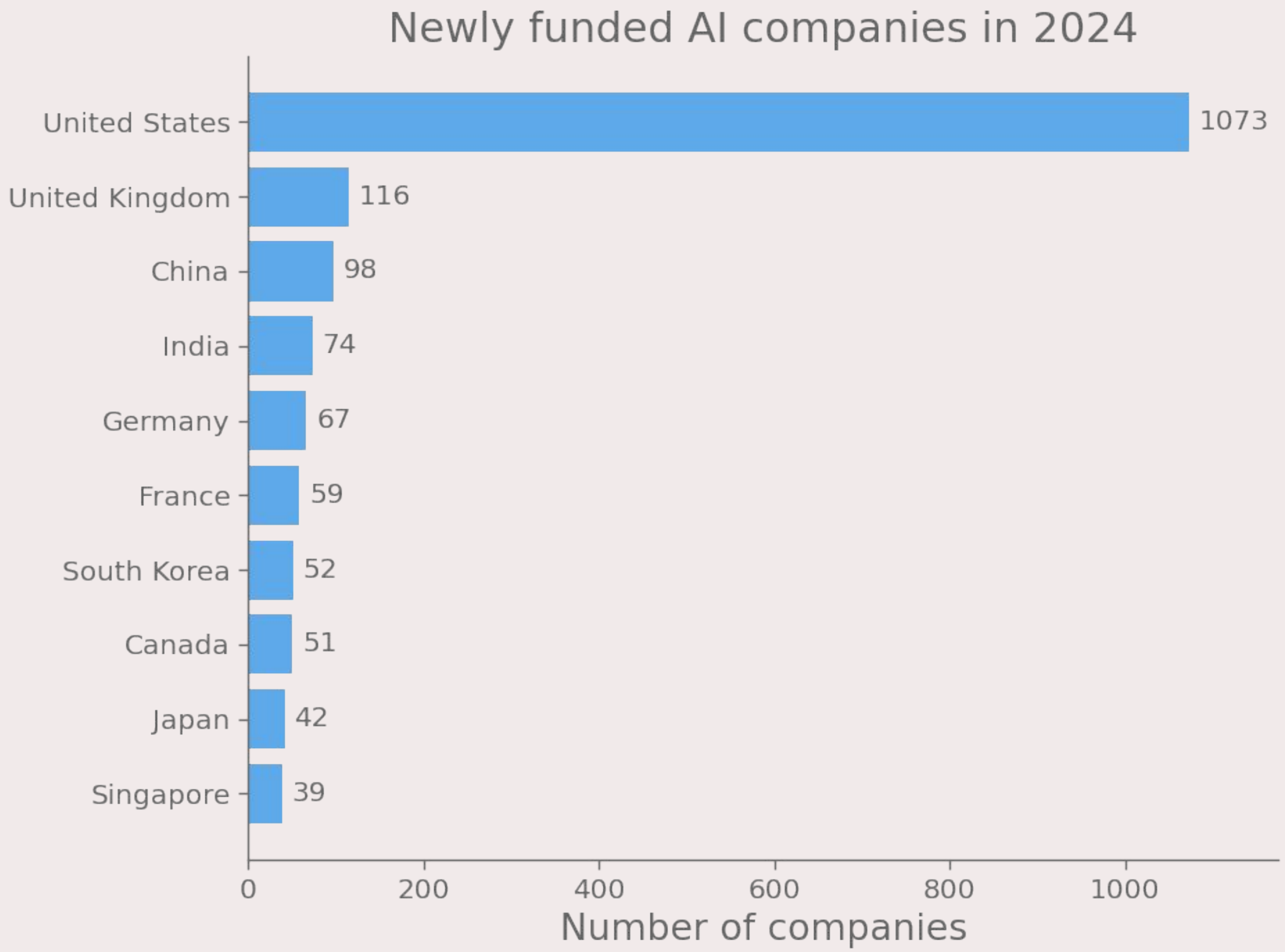


Generative AI investments as percentage of total AI
Percentage
Years
0
2
4
6
8
10
12
14
2014
2016
2018
2020
2022
2024

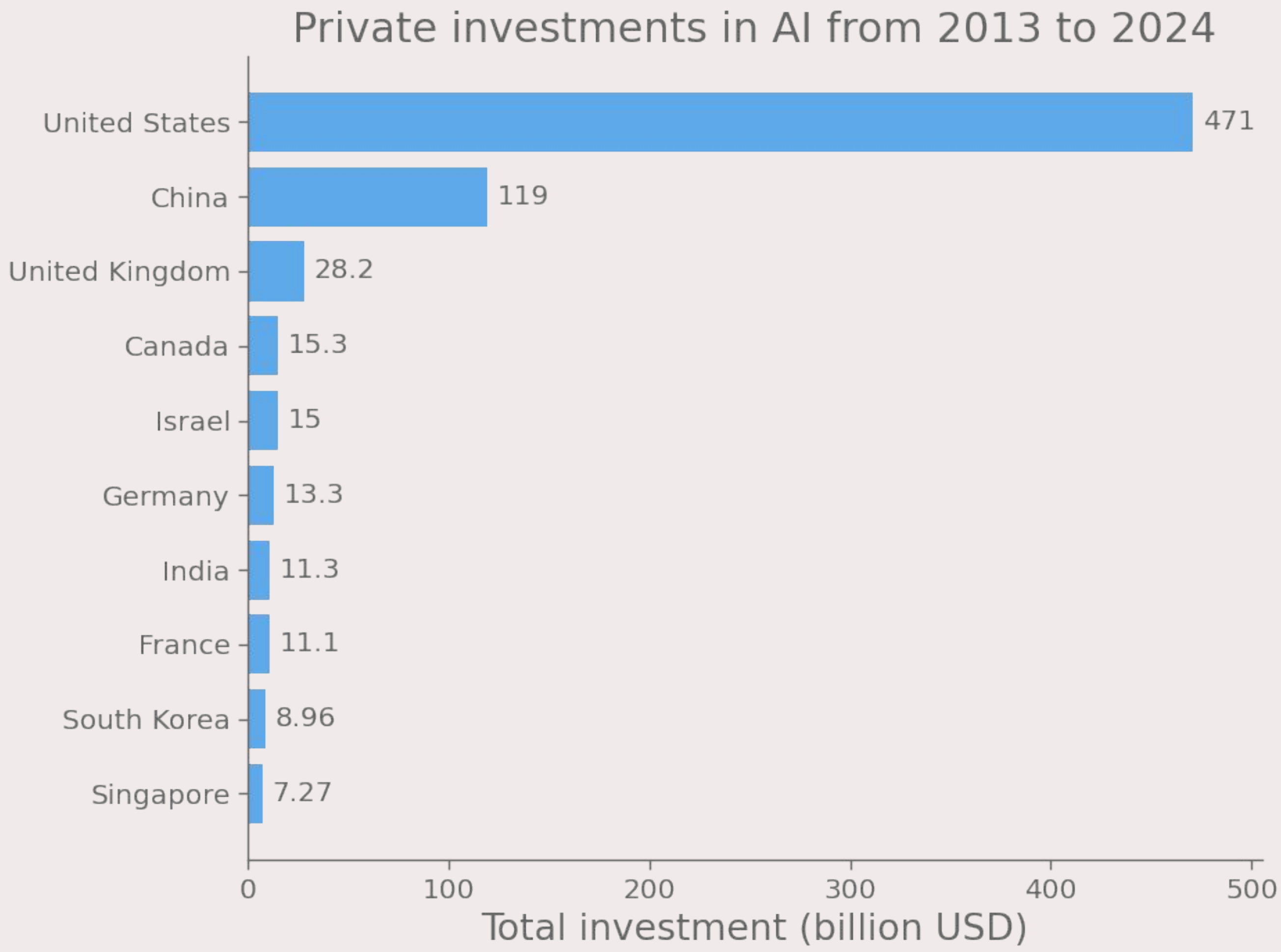


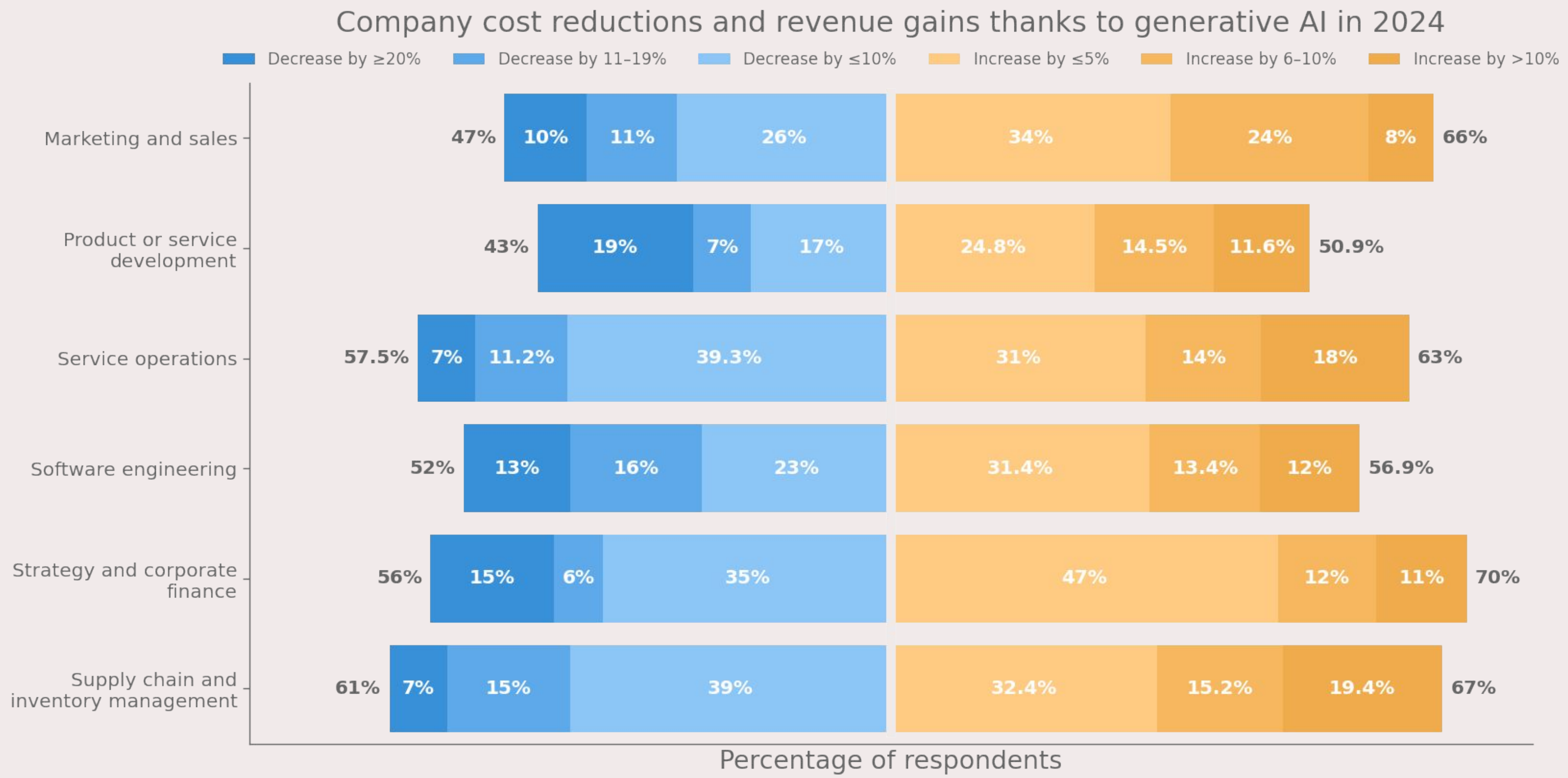
Newly funded AI companies
Generative AI companies
AI companies
Number of companies
0
500
1000
1500
2000
2014
2016
2018
2020
2022
2024
Years
4
6
9
12
18
31
58
93
112
179
214


Generative AI companies as percentage
of newly funded AI companies
Percentage
0
2
4
6
8
10
2014
2016
2018
2020
2022
2024
Years



Newly funded AI companies in 2024
United States
United Kingdom
China
India
Germany
France
South Korea
Canada
Japan
Singapore
1073
116
98
74
67
59
52
51
42
39
0
200
400
600
800
1000
Number of companies

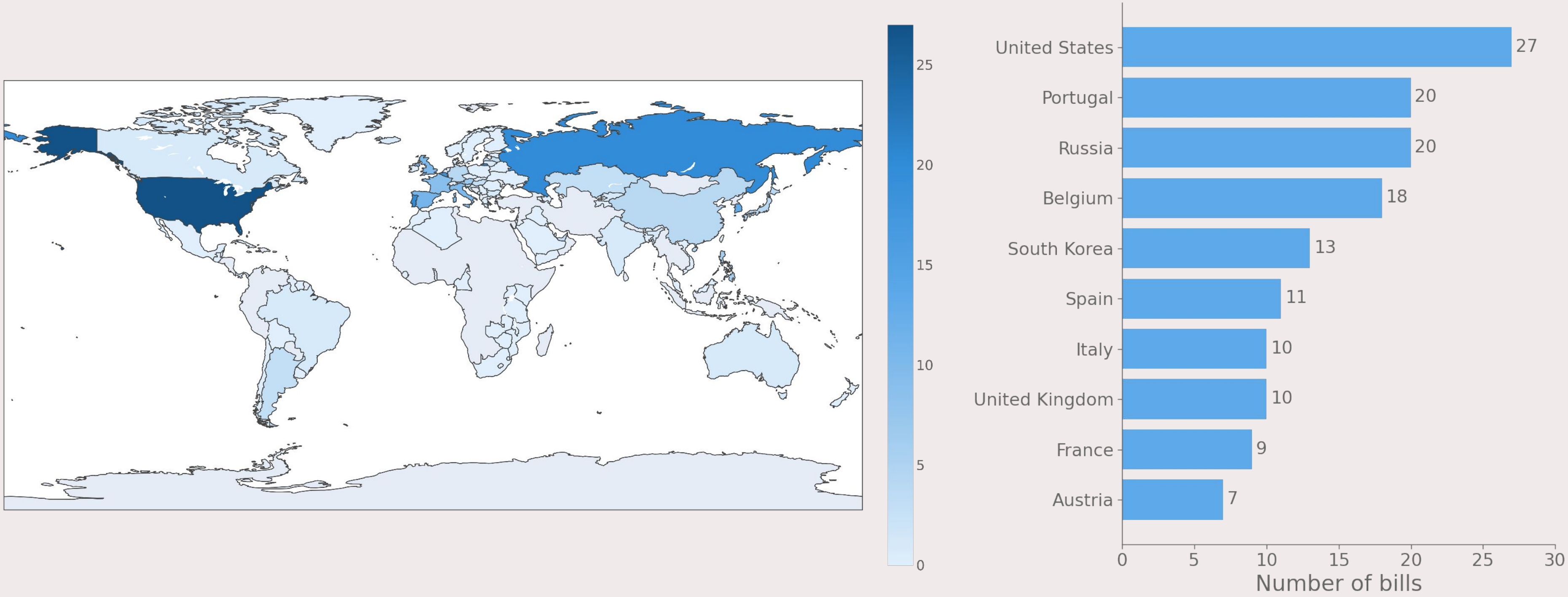

Private investments in AI from 2013 to 2024
United States
China
United Kingdom
Canada
Israel
Germany
India
France
South Korea
Singapore
471
119
28.2
15.3
15
13.3
11.3
11.1
8.96
7.27
0
100
200
300
400
500
Total investment (billion USD)

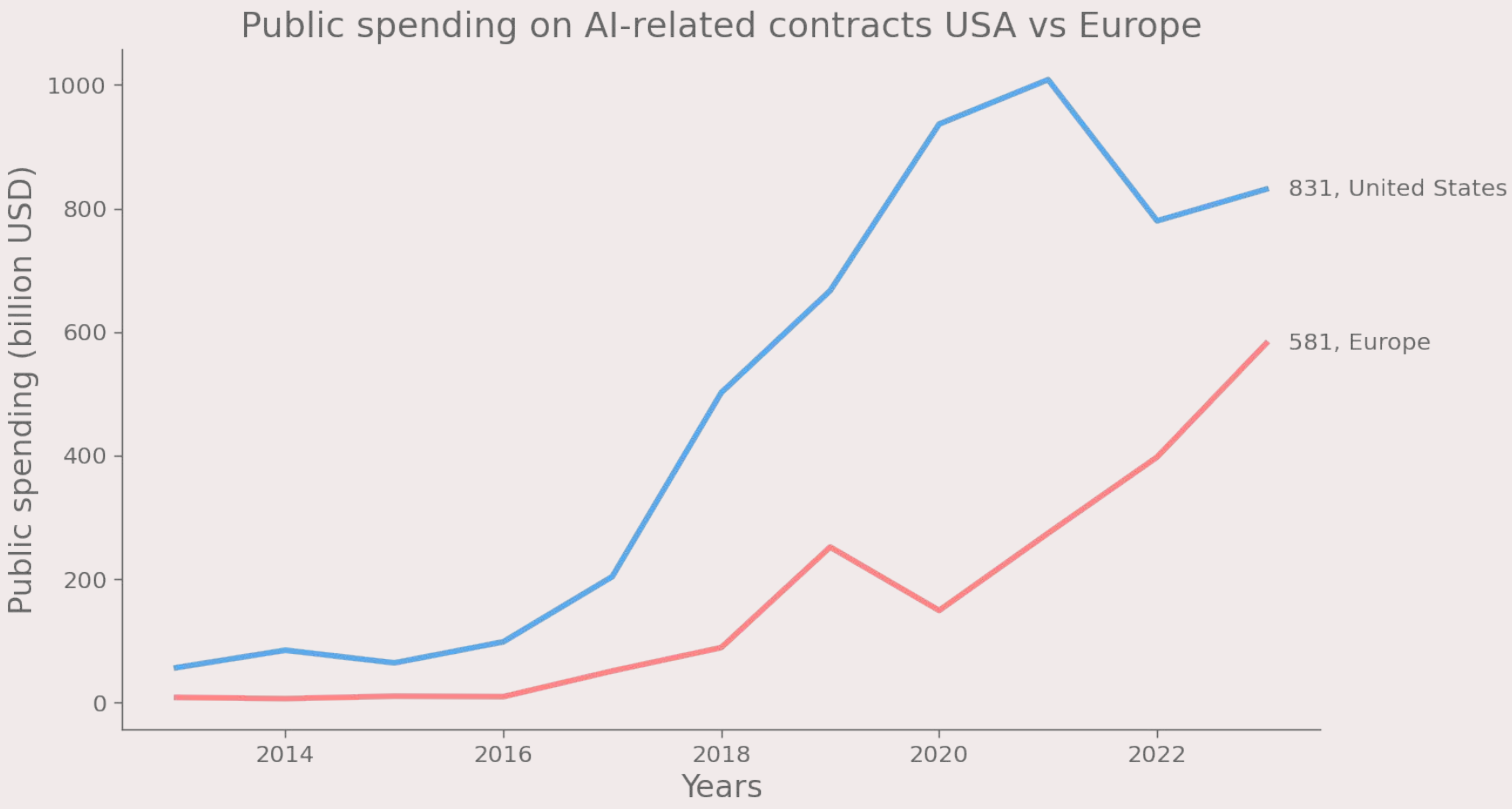


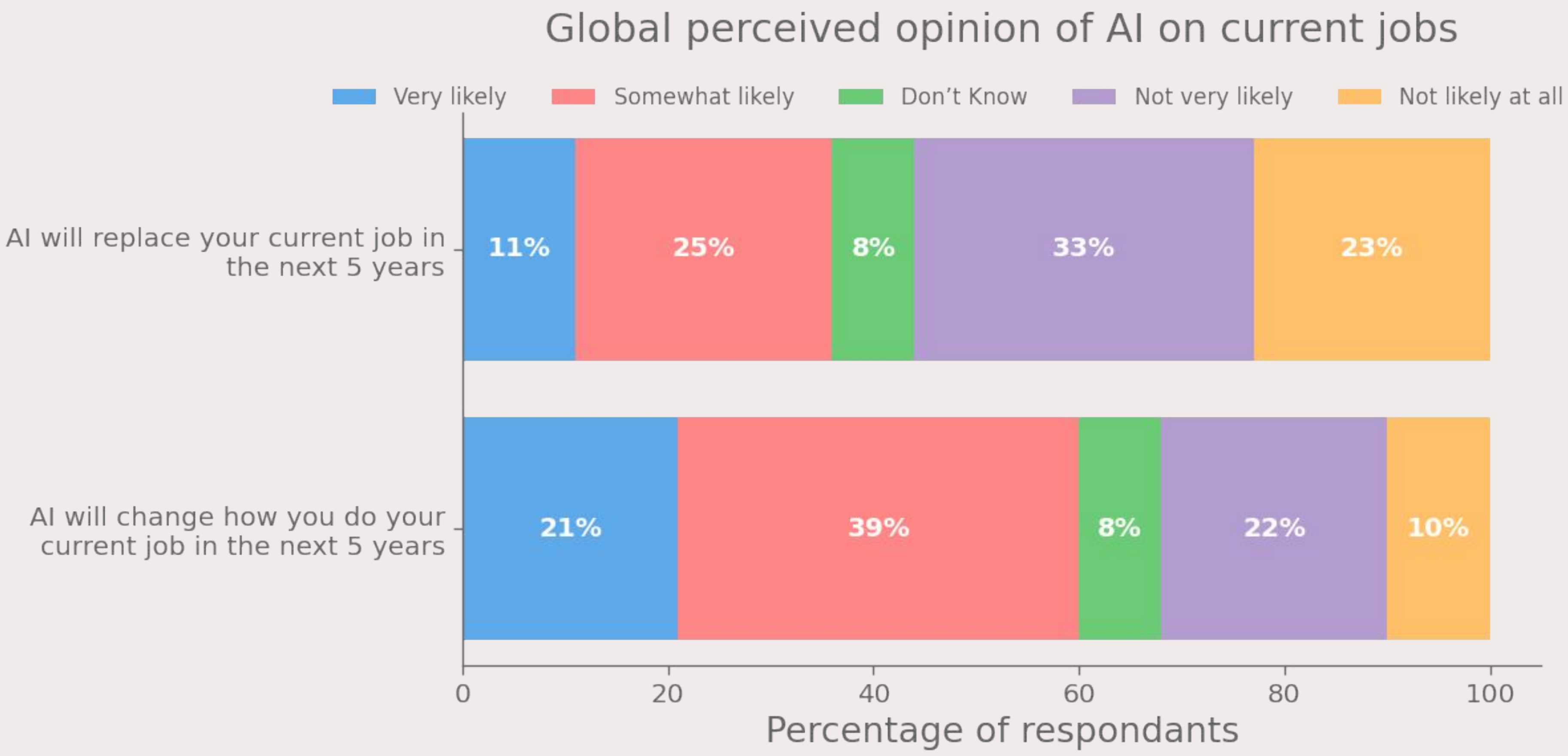
Company cost reductions and revenue gains thanks to generative AI in 2024
Decrease by ≥20%
Decrease by 11–19%
Decrease by ≤10%
Increase by ≤5%
Increase by 6–10%
Increase by >10%
Marketing and sales
47%
10%
11%
26%
34%
24%
8%
66%
Product or service development
43%
19%
7%
17%
24.8%
14.5%
11.6%
50.9%
Service operations
57.5%
7%
11.2%
39.3%
31%
14%
18%
63%
Software engineering
52%
13%
16%
23%
31.4%
13.4%
12%
56.9%
Strategy and corporate finance
56%
15%
6%
35%
47%
12%
11%
70%
Supply chain and inventory management
61%
7%
15%
39%
32.4%
15.2%
19.4%
67%
Percentage of respondents

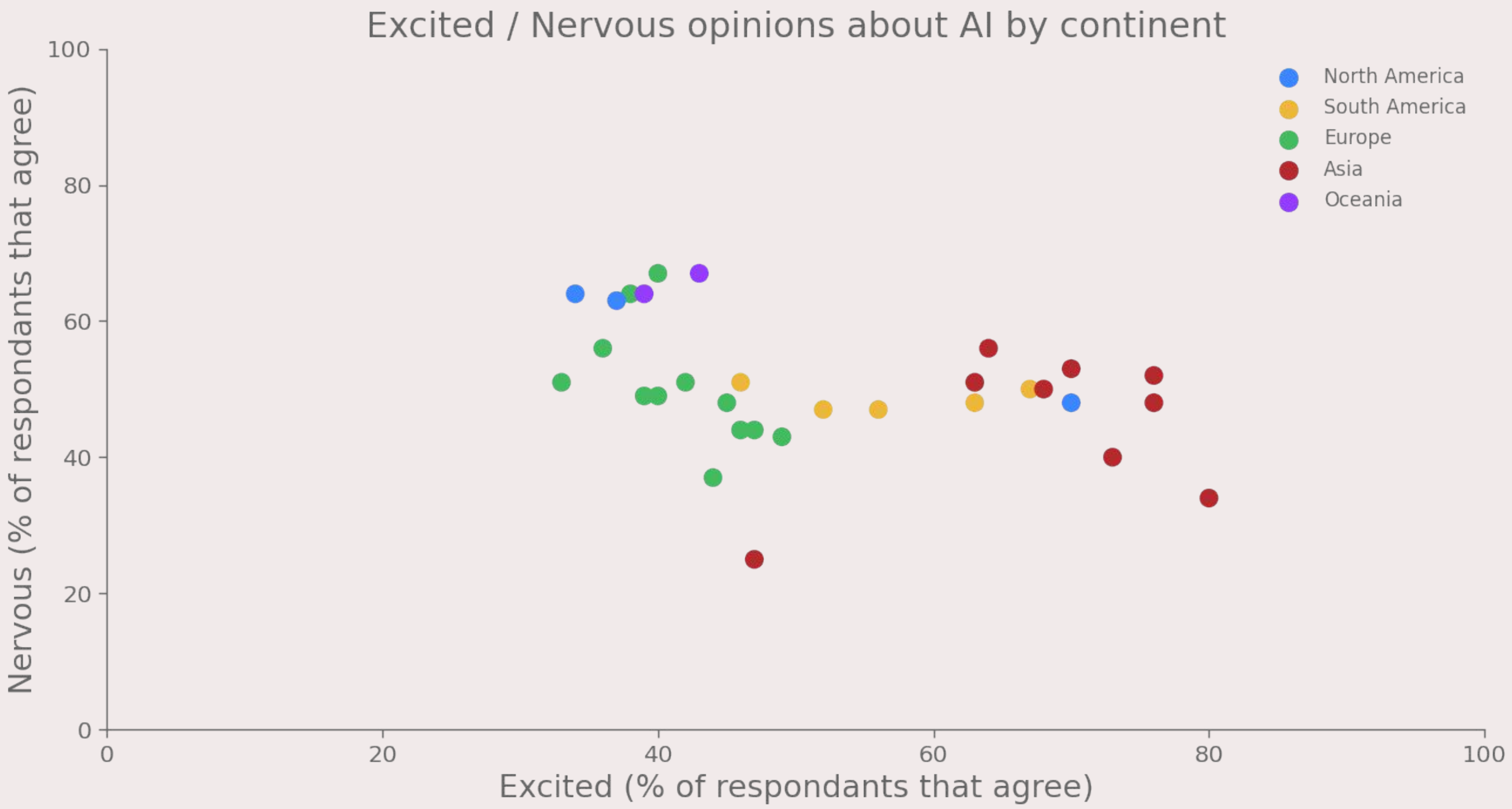
POLICY AND GOVERNANCE

Number of AI-related bills passed into law from 2016 to 2024

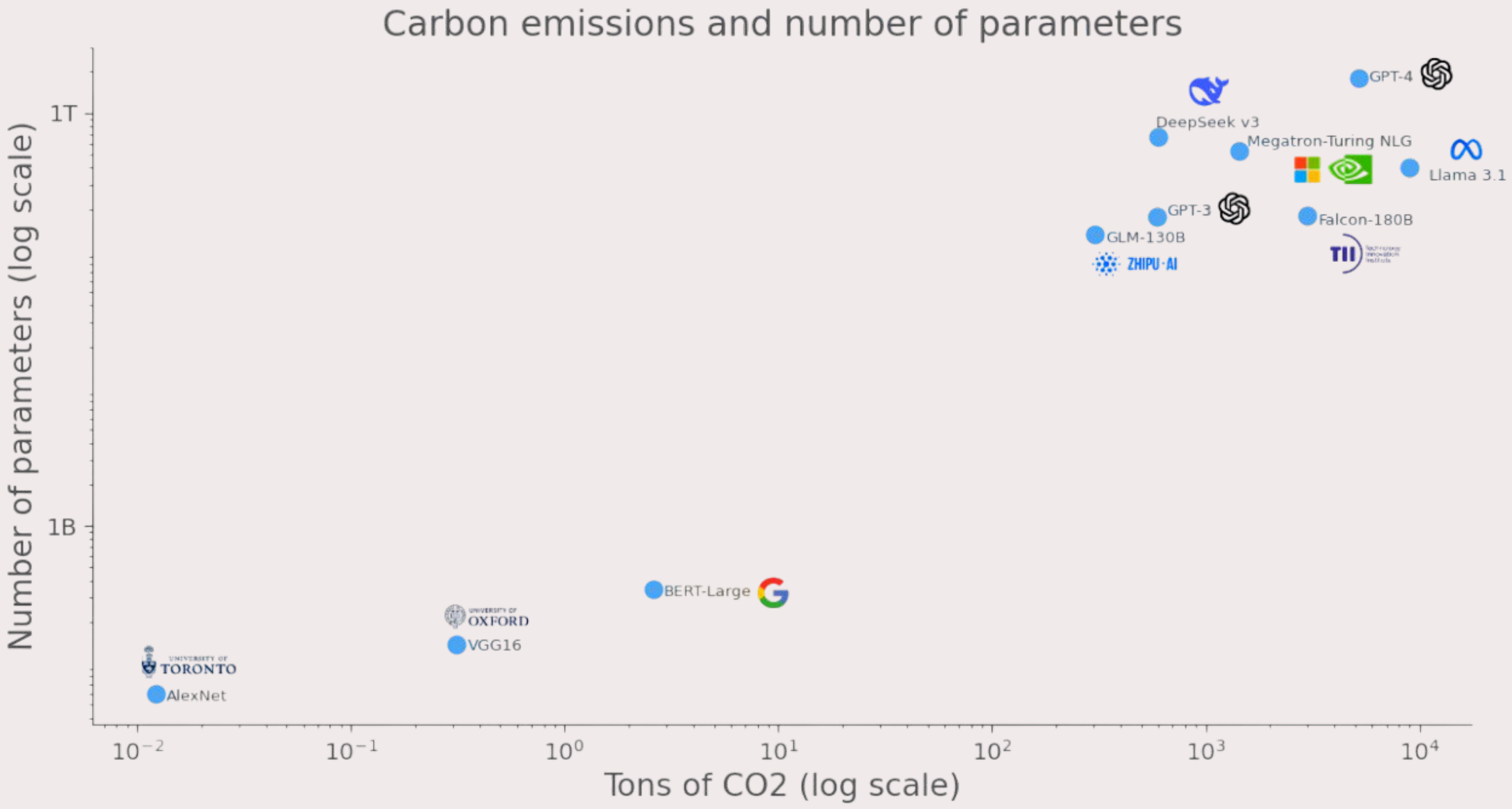



Public spending on AI-related contracts USA vs Europe
Public spending (billion USD)
1000
800
600
400
200
0
2014
2016
2018
2020
2022
Years
831, United States
581, Europe

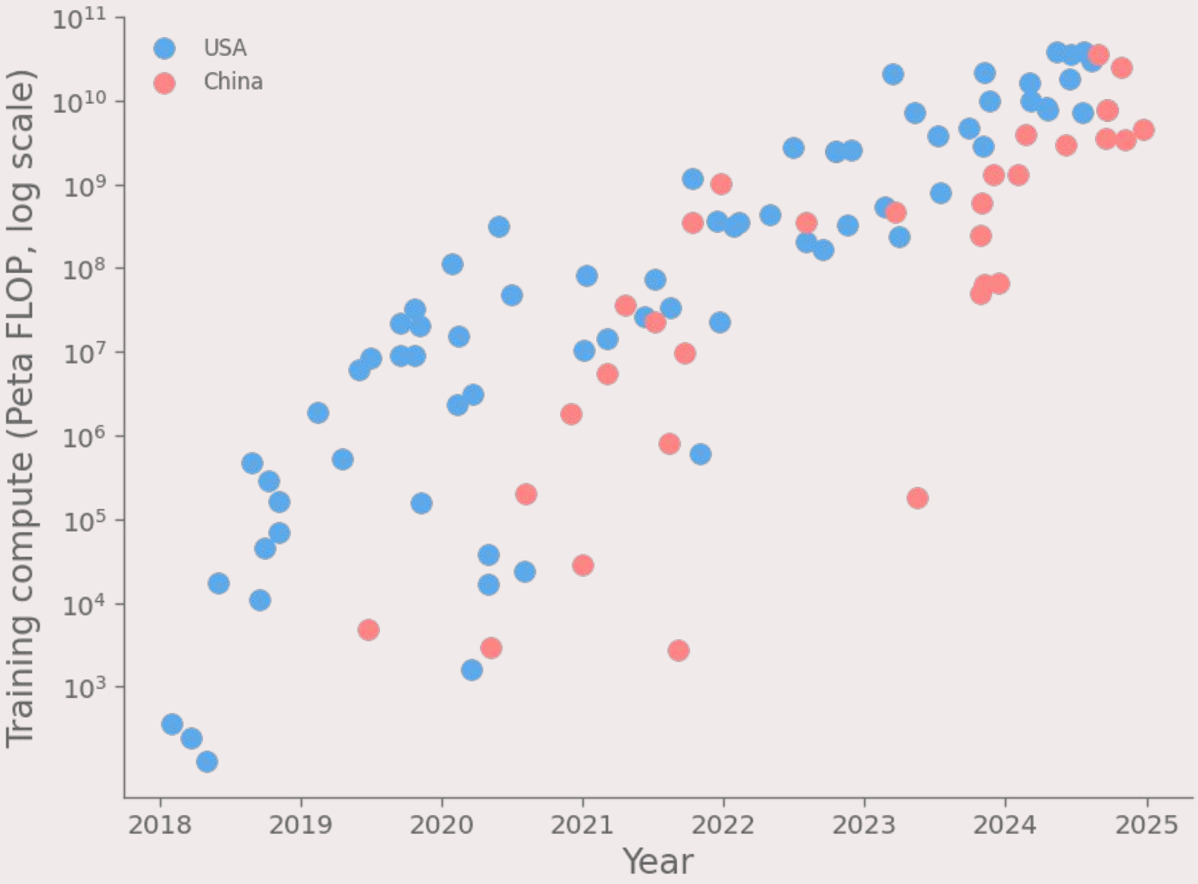

Global perceived opinion of AI on current jobs
Very likely
Somewhat likely
Don’t Know
Not very likely
Not likely at all
AI will replace your current job in the next 5 years
11%
25%
8%
33%
23%
AI will change how you do your current job in the next 5 years
21%
39%
8%
22%
10%
0
20
40
60
80
100
Percentage of respondants

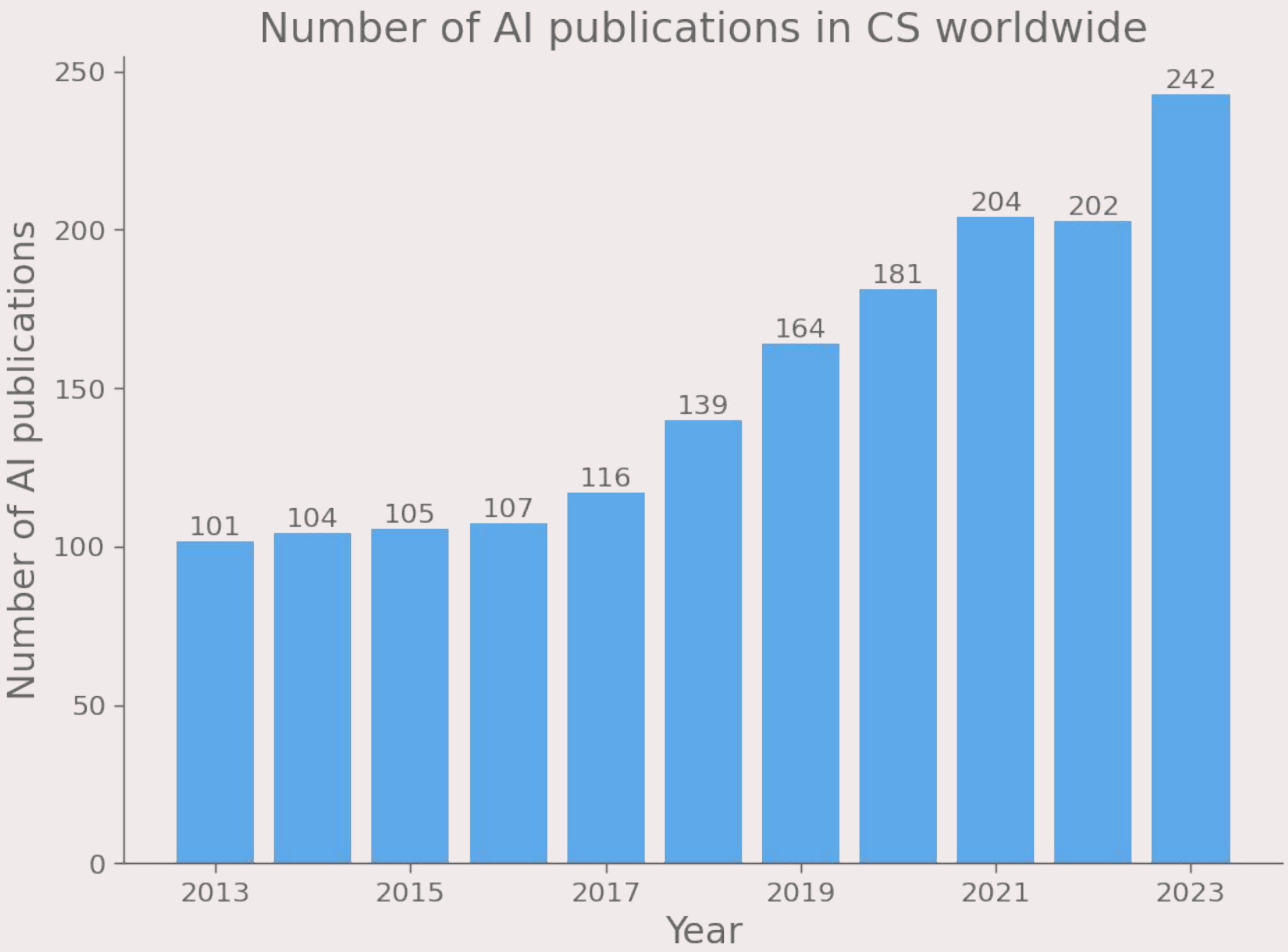

Excited / Nervous opinions about AI by continent
North America
South America
Europe
Asia
Oceania
Nervous (% of respondants that agree)
Excited (% of respondants that agree)
0
20
40
60
80
100

RESEARCH AND DEVELOPMENT

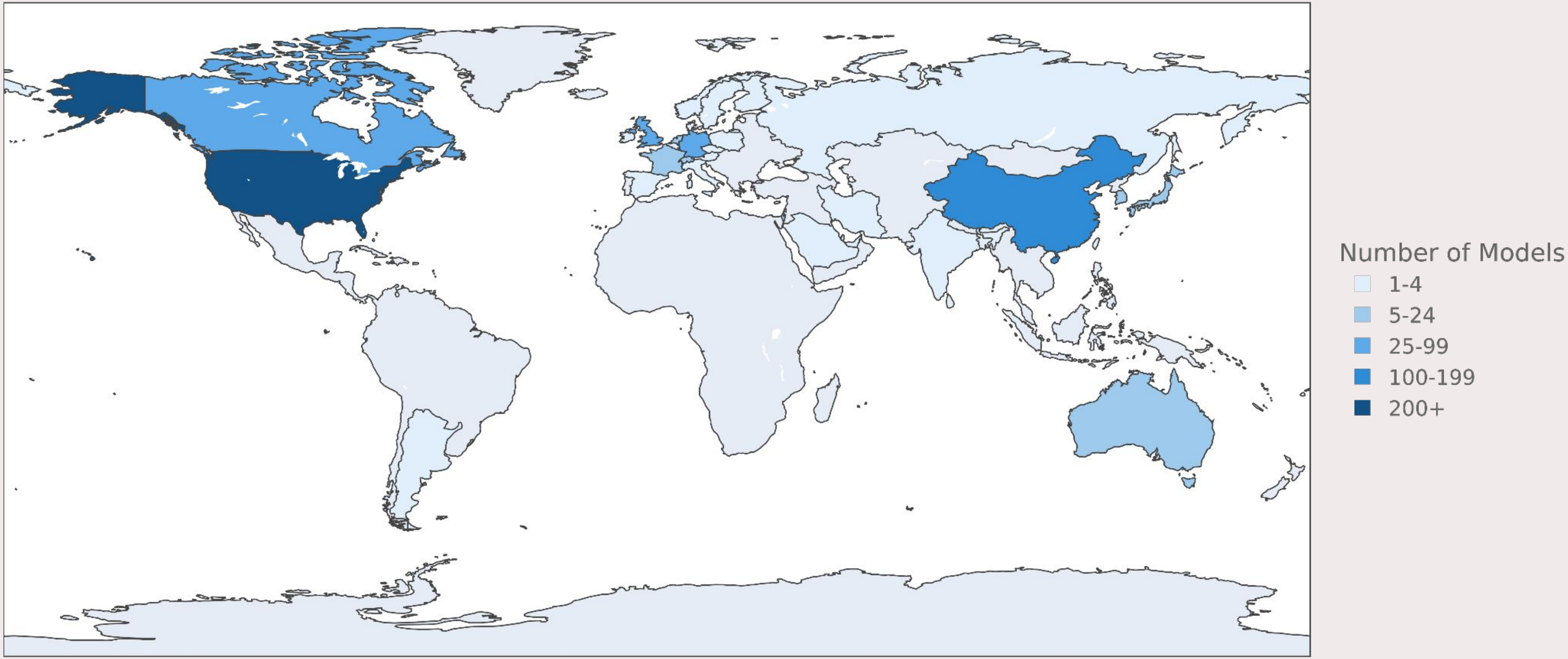

Carbon emissions and number of parameters
Number of parameters (log scale)
1T
1B
Tons of CO2 (log scale)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
GPT-4
DeepSeek v3
Megatron-Turing NLG
Llama 3.1
GPT-3
Falcon-180B
GLM-130B
ZHIPU·AI
TII
BERT-Large
UNIVERSITY OF OXFORD
VGG16
UNIVERSITY OF TORONTO
AlexNet

Training compute of select notable AI models in the United States and China

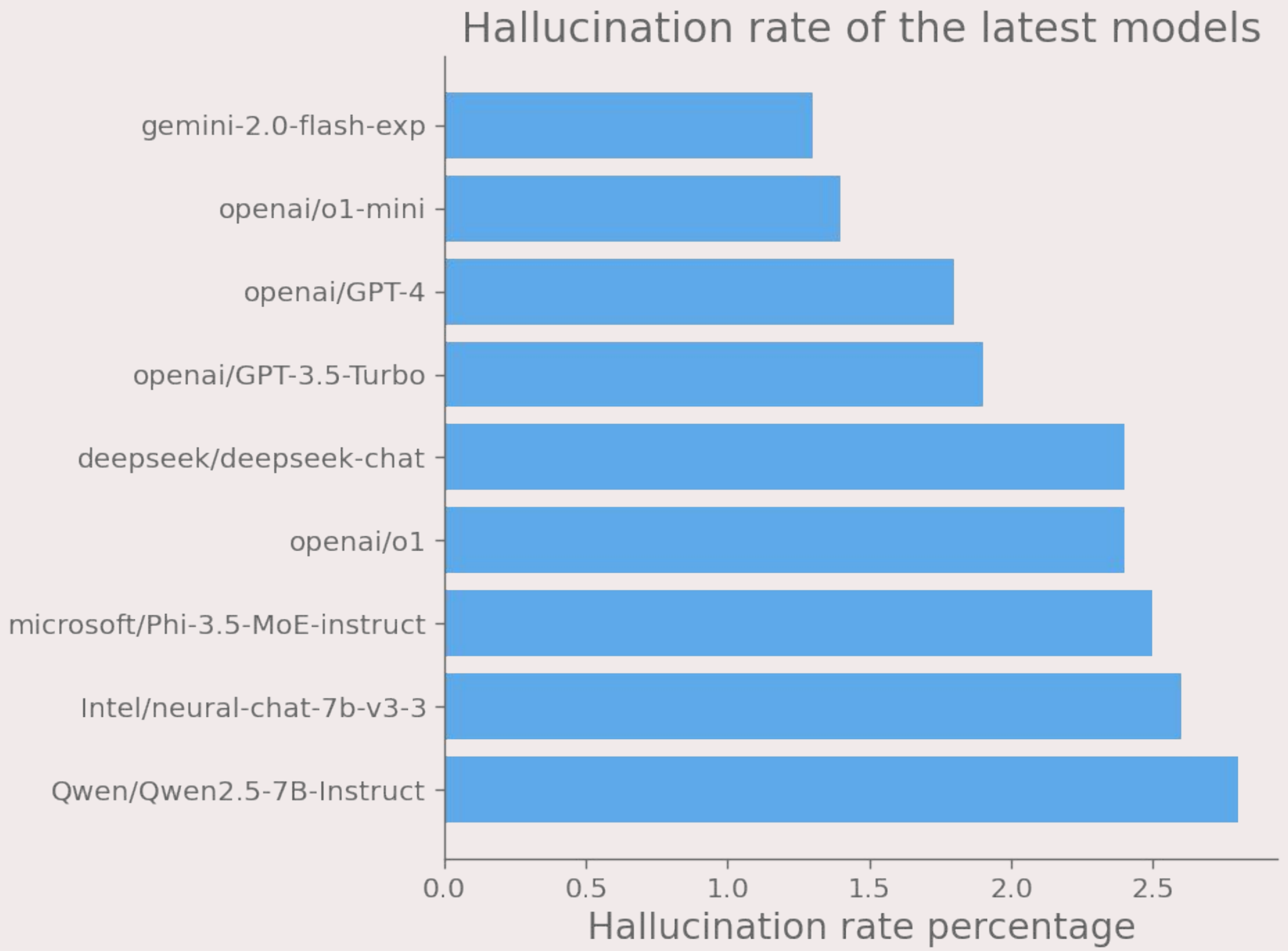


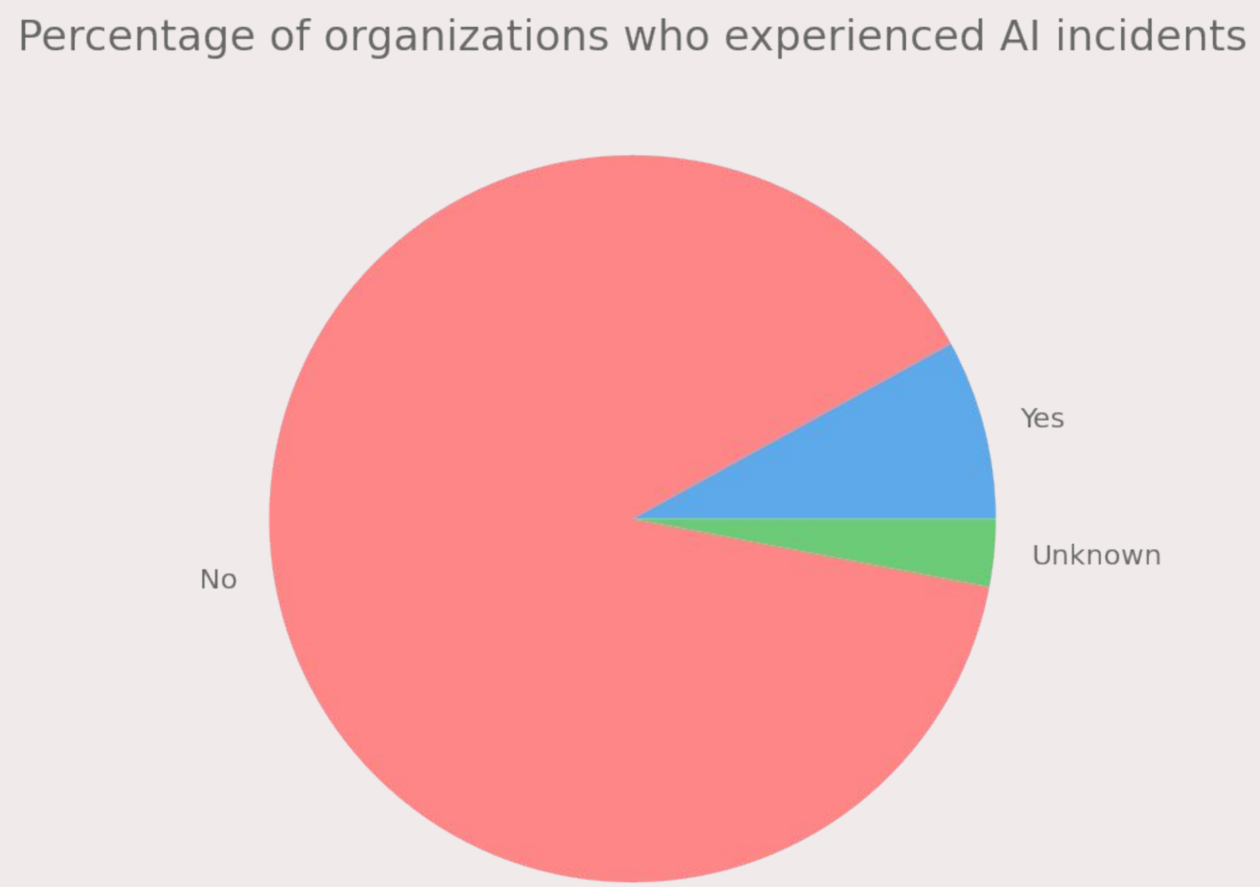

Number of AI publications in CS worldwide
Number of AI publications
250
200
150
100
50
0
101
104
105
107
116
139
164
181
204
202
242
2013
2015
2017
2019
2021
2023
Year

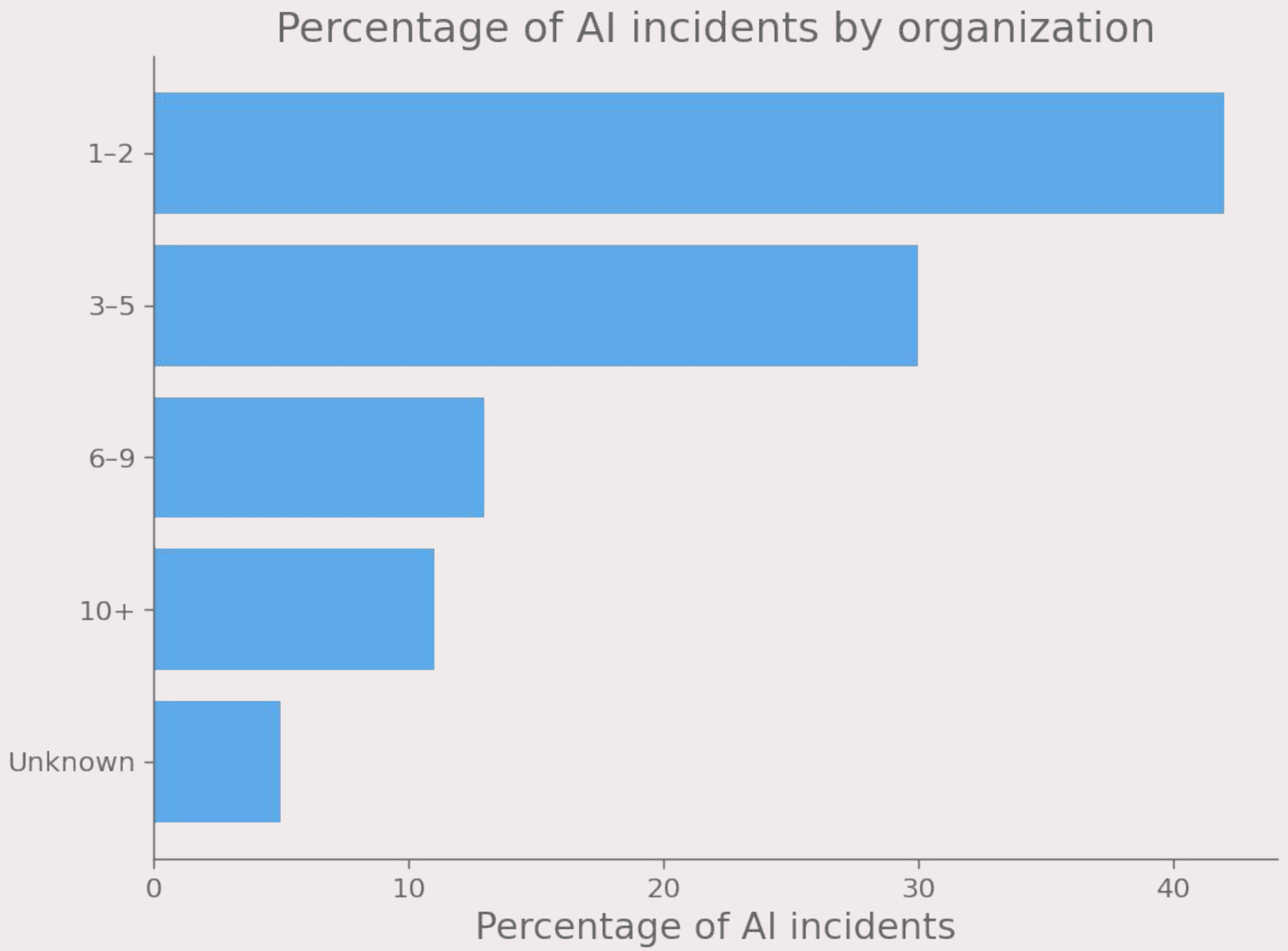
Number of Notable AI models from 2003 to 2024



RESPONSIBLE AI

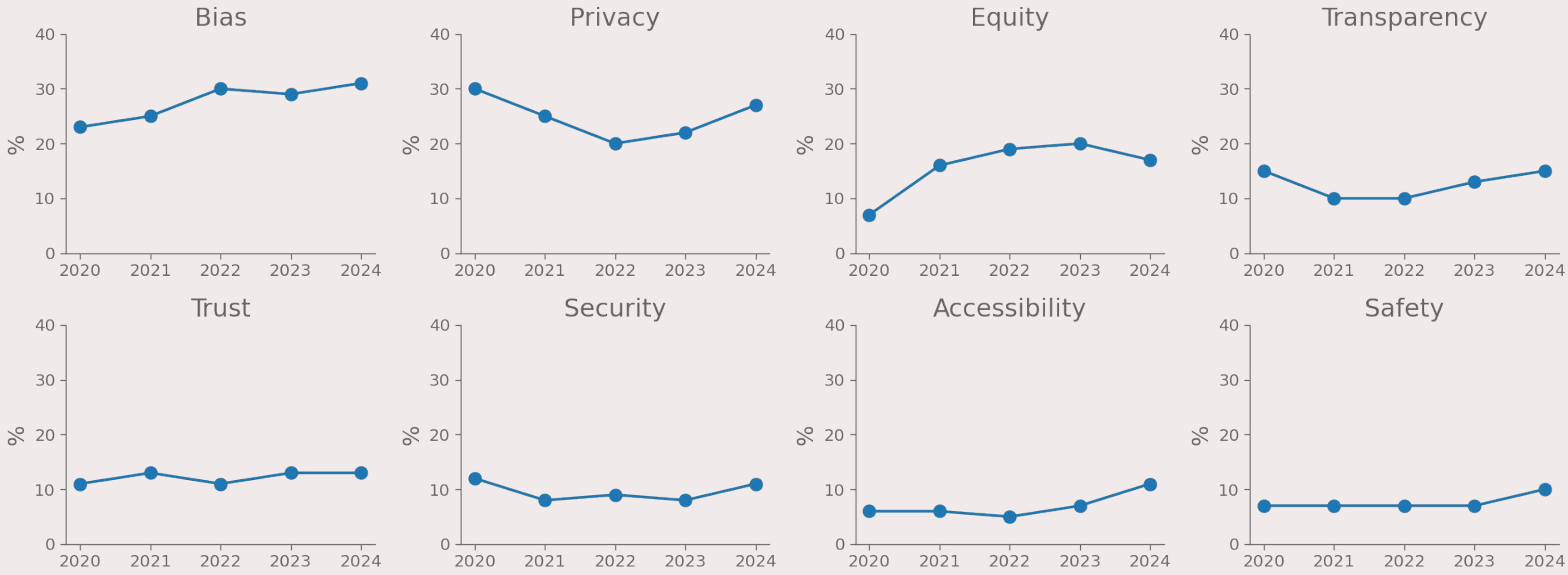

Hallucination rate of the latest models
gemini-2.0-flash-exp
openai/o1-mini
openai/GPT-4
openai/GPT-3.5-Turbo
deepseek/deepseek-chat
openai/o1
microsoft/Phi-3.5-MoE-instruct
Intel/neural-chat-7b-v3-3
Qwen/Qwen2.5-7B-Instruct
0.0
0.5
1.0
1.5
2.0
2.5
Hallucination rate percentage

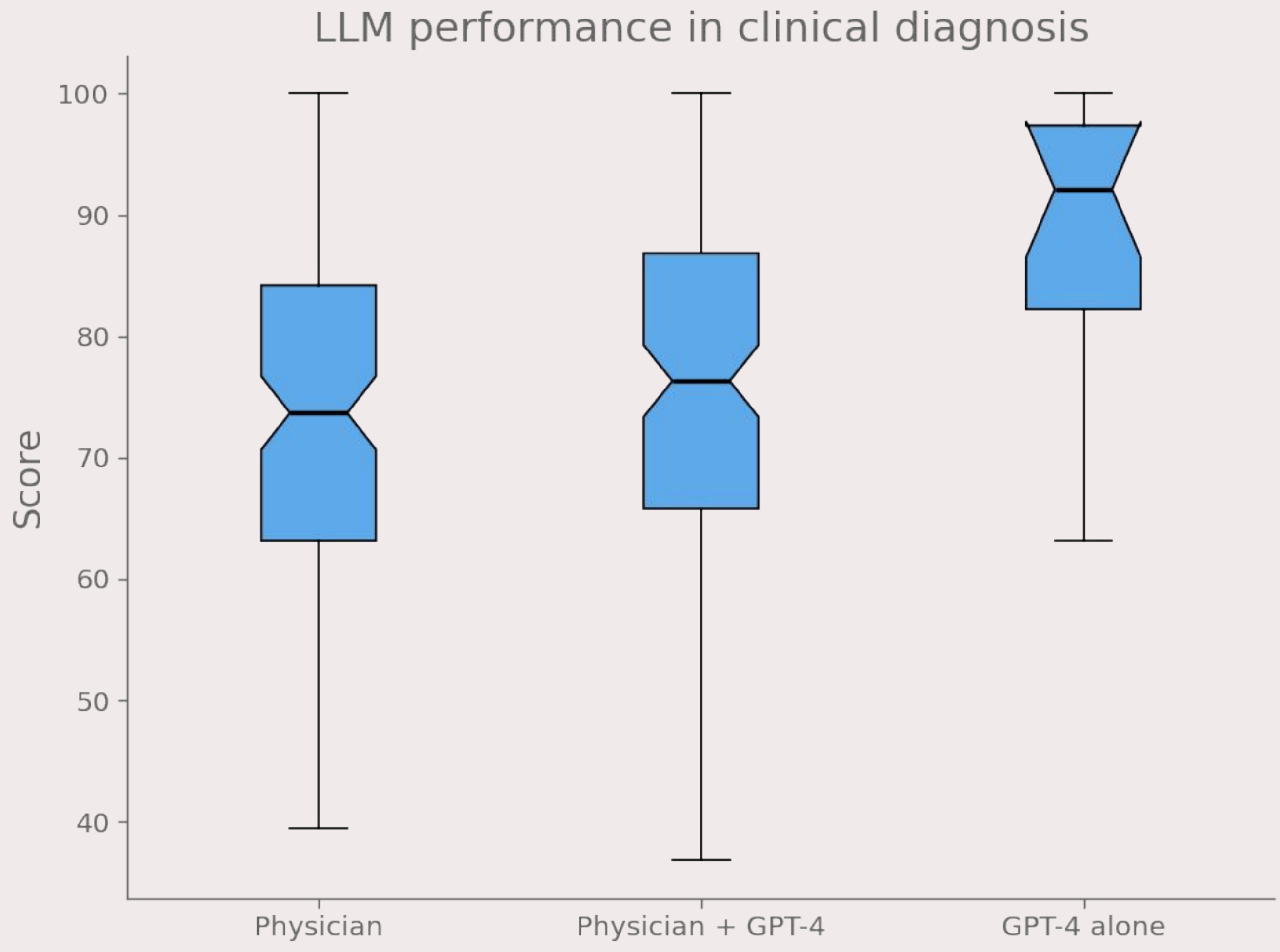

Percentage of organizations who experienced AI incidents
Yes
Unknown
No


Percentage of AI incidents by organization
1–2
3–5
6–9
10+
Unknown
0
10
20
30
40
Percentage of AI incidents

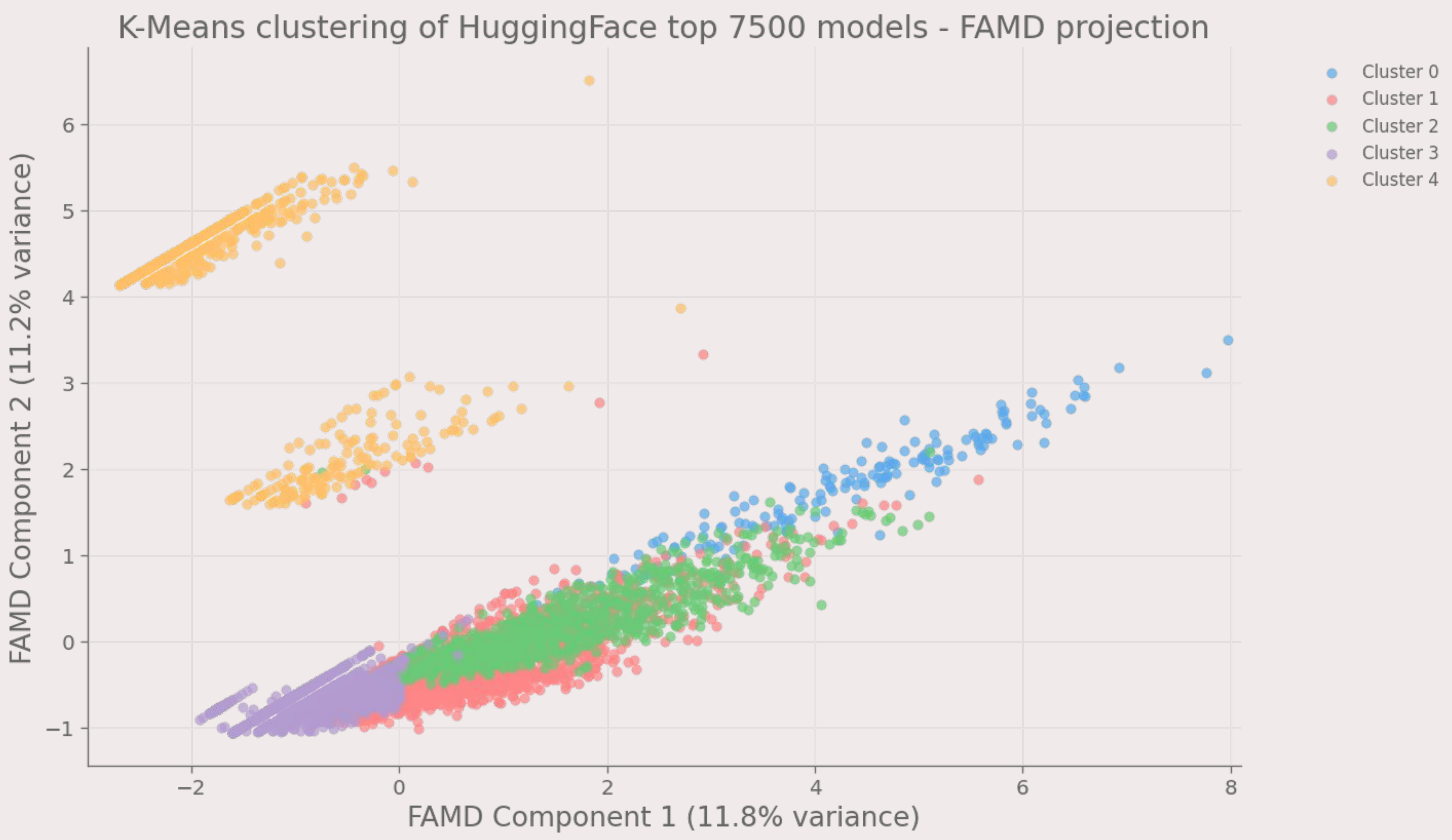
SCIENCE AND MEDICINE

Top 8 ethical concerns discussed in medical AI ethics publications



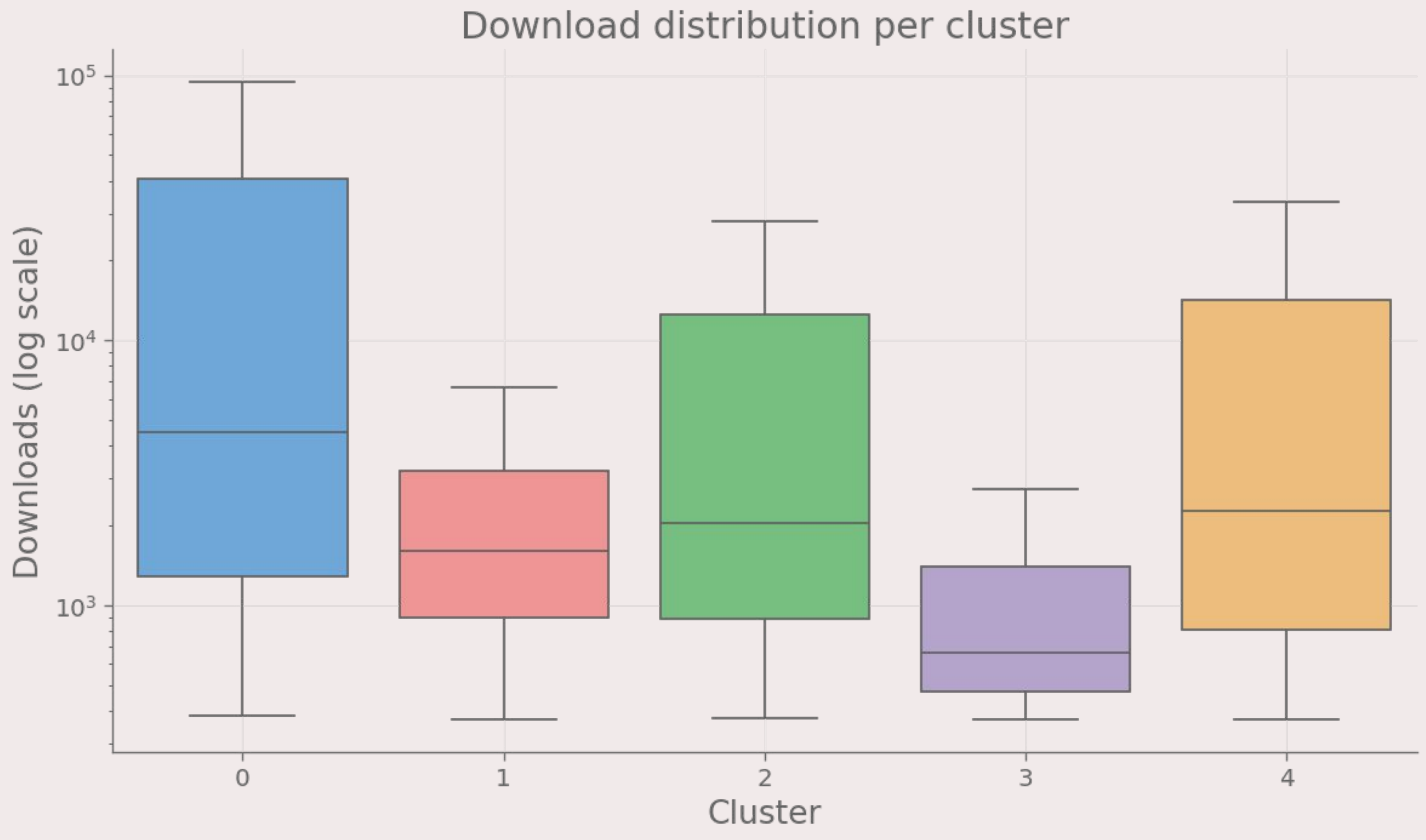

LLM performance in clinical diagnosis
Score
100
90
80
70
60
50
40
Physician
Physician + GPT-4
GPT-4 alone

CLUSTERING

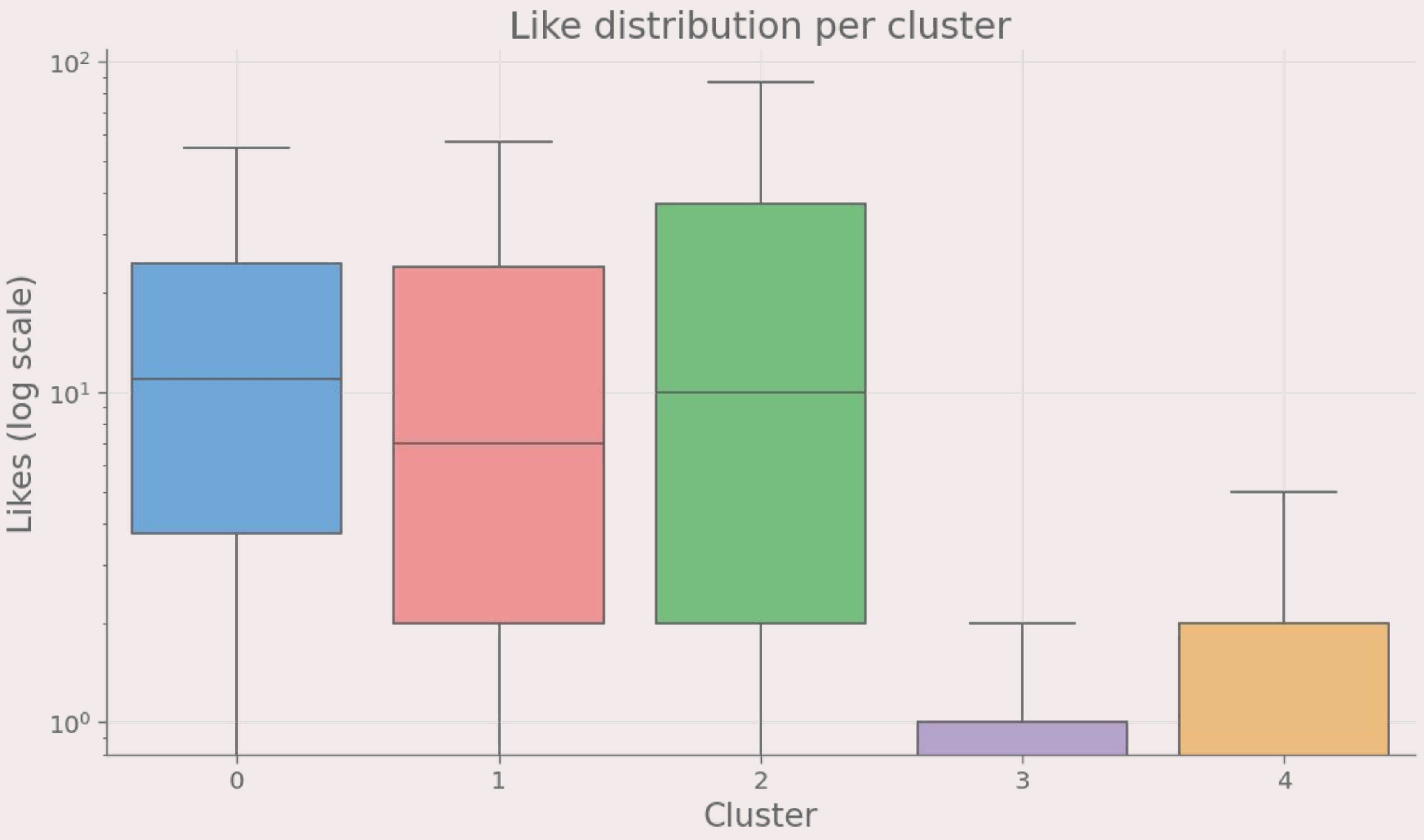

K-Means clustering of HuggingFace top 7500 models - FAMD projection
FAMD Component 2 (11.2% variance)
FAMD Component 1 (11.8% variance)
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
−2
0
2
4
6
8
−1
0
1
2
3
4
5
6

Cluster	Size	Avg. Downloads	Avg. Likes	Top Architecture	Top Task
0 (🔵)	176	746.512	59	sentence-transformers	sentence-similarity
1 (🔴)	3.319	54.139	41	transformers	text generation
2 (🟢)	894	169.927	56	transformers	other
3 (🟣)	2.456	1.215	2	transformers	unknown
4 (🟠)	655	277.425	3	timm	image classification

Cluster	Downloads			Likes		
	Average	Median	SD	Average	Median	SD
0 ()	746.512	4.522	7.316.683	59	11	318
1 ()	54.139	1.608	1.032.386	41	7	147
2 ()	169.927	2.042	1.511.338	56	10	191
3 ()	1.215	665	1.822	2	0	4
4 ()	277.425	2.266	4.781.000	3	0	7



Download distribution per cluster
Downloads (log scale)
10^5
10^4
10^3
0
1
2
3
4
Cluster



Like distribution per cluster
Likes (log scale)
10^2
10^1
10^0
0
1
2
3
4
Cluster

Cluster	Wilcoxon's Test	
	Downloads	Likes
0 (🔵)	5.4e-20	2.5e-17
1 (🔴)	4.3e-52	1.7e-239
2 (🟢)	3.3e-37	2.6e-82
3 (🟣)	1.1e-261	0
4 (🟠)	4.68e-29	7.7e-52

Feature	Contributes to FAMD dimensions		
	Dimension 1	Dimension 2	Total
Architecture	0.319	0.464	0.784
Task	0.226	0.447	0.675
Likes	0.281	0.000	0.281
Downloads	0.173	0.088	0.261

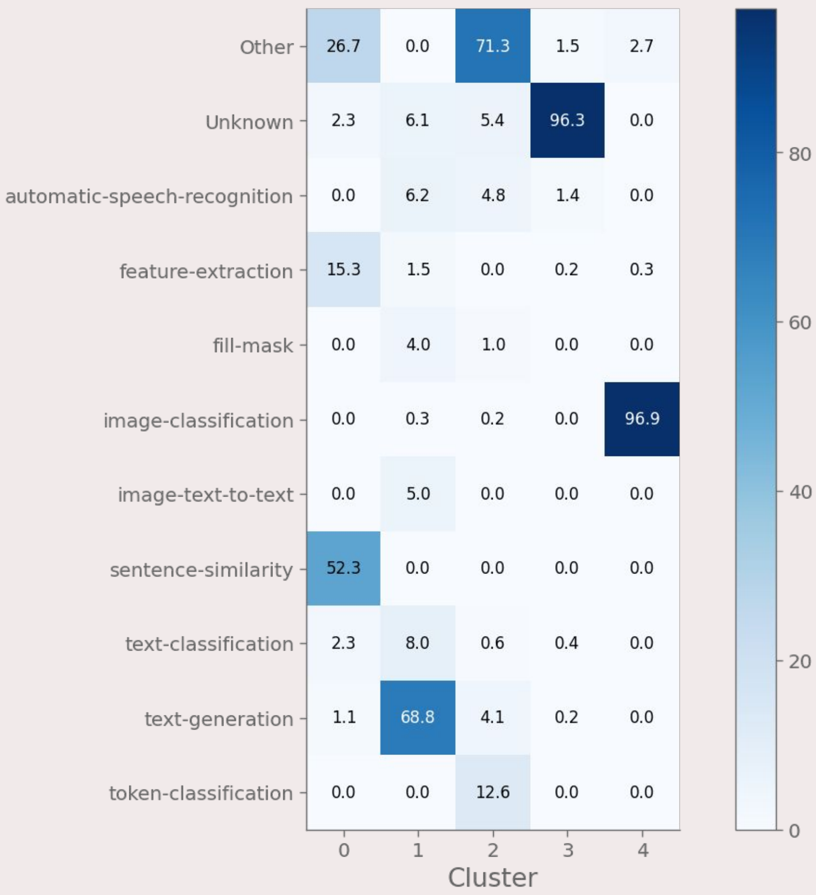
Percentage distribution of tasks per cluster

Task	Clusters				
	CL 0	CL 1	CL 2	CL 3	CL 4
automatic speech recognition	0	6.2	4.8	1.4	0
feature extraction	15.3	1.5	0	0.2	0.3
text generation	1.1	68.8	4.1	0.2	0
image classification	0	0.3	0.2	0	96.9
token classification	2.3	8	0.6	0.4	0
sentence similarity	52.3	0	0	0	0
text classification	2.3	8	0.6	0.4	0
other	26.7	0	71.3	1.5	2.7
unknown	2.3	6.1	5.4	96.3	0

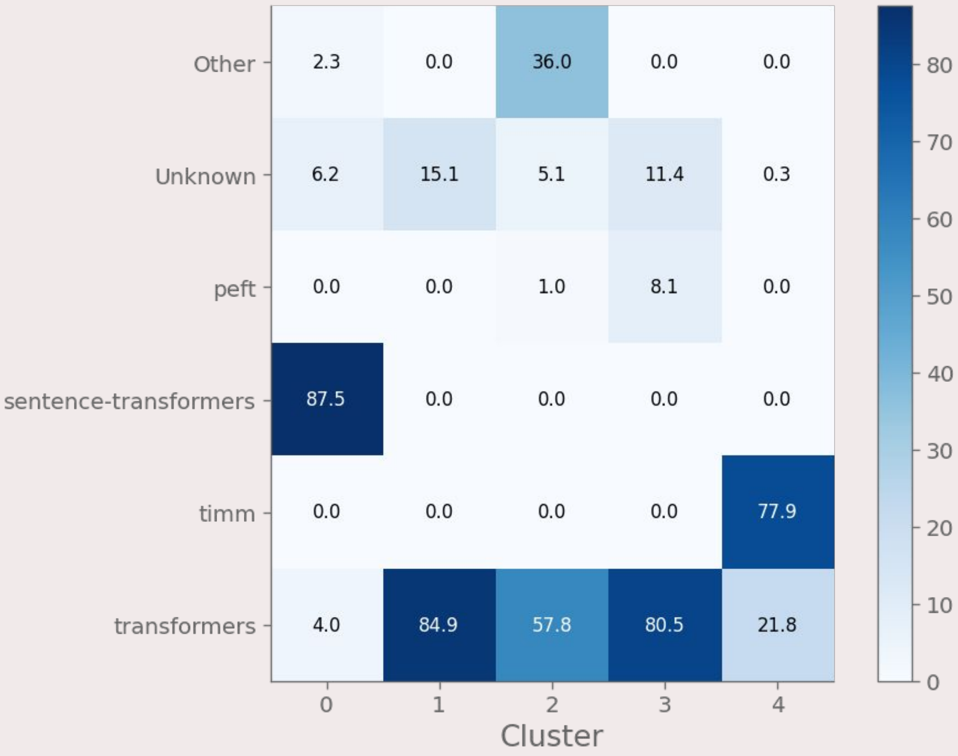
Percentage distribution of architectures per cluster

Architecture	Clusters				
	CL 0	CL 1	CL 2	CL 3	CL 4
peft	0	0	1	8.1	0
sentence transformers	87.5	0	0	0	0
timm	0	0	0	0	77.9
transformers	4	84.9	57.8	80.5	21.8
other	2.3	0	36	0	0
unknown	6.2	15.1	5.1	11.4	0.3

Percentage of tasks per cluster



Percentage of architectures per cluster



SOURCES

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THANKS FOR THE ATTENTION